

Formulas

set of classes:	C
set of samples:	S
feature vectors:	$x_s \in \mathbb{R}^n$
weights:	$\theta_c \in \mathbb{R}^n$
labels:	$y_{s,c} \in \{0, 1\}$
loss function:	$L(x, y) = -xy + \ln(1 + e^x)$
regularizer:	$R(\theta_c) = \frac{\lambda}{2} \ \theta_c\ ^2$
goal:	$\min \sum_{c \in C} R(\theta_c) + \sum_{s \in S} L(\langle \theta_c, x_s \rangle, y_{s,c})$
constraints:	$\forall s \in S : \sum_{c \in C} y_{s,c} = 1$
	$\forall c \in C : \sum_{s \in S} y_{s,c} = \frac{ S }{ C }$

ILP

$$\begin{aligned}
 &\text{given } \theta_c \text{ and } x_s \\
 &\min \sum_{c \in C} R(\theta_c) + \sum_{s \in S} L(\langle \theta_c, x_s \rangle, y_{s,c}) \\
 &\min \sum_{c \in C} \sum_{s \in S} -\langle \theta_c, x_s \rangle y_{s,c}
 \end{aligned}$$

LP

$$\begin{aligned}
 &\text{given } y_{s,c} \text{ and } x_s \\
 &\min \sum_{s \in S} \sum_{c \in C} (1 - y_{sc}) \mu_{sc}^+ + y_{sc} \nu_{sc}^+ \\
 &\text{subject to } \forall s \in S, c \in C : \\
 &\quad a + \langle \theta_c, x_s \rangle = \mu_{sc}^+ - \mu_{sc}^- \\
 &\quad a - \langle \theta_c, x_s \rangle = \nu_{sc}^+ - \nu_{sc}^- \\
 &\quad \mu_{sc}^+, \mu_{sc}^-, \nu_{sc}^+, \nu_{sc}^- \geq 0 \\
 &\min \sum_{s \in S} \sum_{c \in C} z_{sc} \\
 &\text{subject to } \forall s \in S, c \in C : \\
 &\quad z_{sc} \geq 0 \\
 &\quad y_{sc} \Rightarrow a + \langle \theta_c, x_s \rangle \leq z_{sc} \\
 &\quad \neg y_{sc} \Rightarrow a - \langle \theta_c, x_s \rangle \leq z_{sc}
 \end{aligned}$$

GD

given $y_{s,c}$ and x_s

$$\forall c \in C : \nabla_{\theta_c} \left(R(\theta_c) + \sum_{s \in S} L(\langle \theta_c, x_s \rangle, y_{s,c}) \right) = \lambda \theta_c + \sum_{s \in S} x_s \left(-y_{s,c} + \frac{1}{e^{-\langle \theta_c, x_s \rangle} + 1} \right)$$